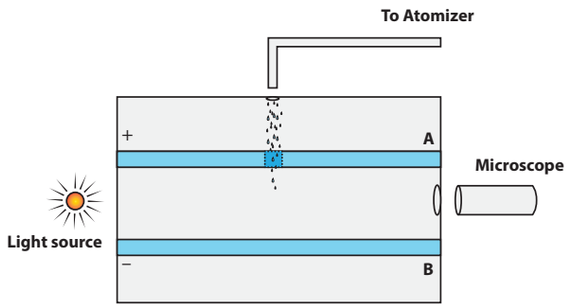


(a)



(b)

Figure 9.6 Millikan's experiment (a) real picture and schematic picture (b) Side view picture

plates are enclosed in a chamber with glass walls. Further, plates A and B are maintained at high potential difference around 10 kV such that electric field acts vertically downward. A small hole is made at the centre of the upper plate A and an atomizer is kept exactly above the hole to spray the liquid. When a fine droplet of the highly viscous non volatile liquid (like glycerine) is sprayed using atomizer, they fall freely downward through the hole of the top plate only under the influence of gravity.

Few oil drops in the chamber can acquire electric charge (negative charge) because of friction with air or passage of x-rays in between the parallel plates. Further the chamber is illuminated by light which is passed horizontally and oil drops can be seen clearly using microscope placed perpendicular to the light beam. These drops can move either upwards or downward.

Let m be the mass of the oil drop and q be its charge. Then the forces acting on the droplet are

(a) gravitational force $F_g = mg$
(b) electric force $F_e = qE$
(c) buoyant force F_b
(d) viscous force F_v

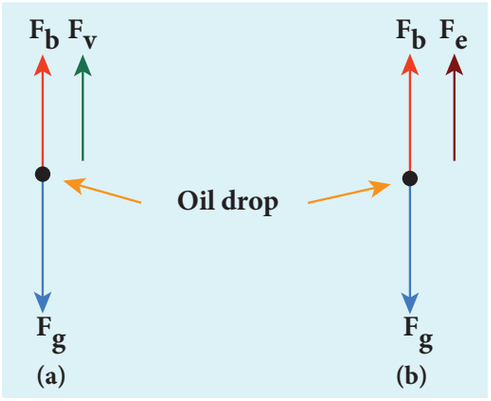


Figure 9.7 Free body diagram of the oil drop – (a) without electric field (b) with electric field

(a) Determination of radius of the droplet

When the electric field is switched off, the oil drop accelerates downwards. Due to the presence of air drag forces, the oil drops easily attain its terminal velocity and moves with constant velocity. This velocity can be carefully measured by noting down the time taken by the oil drop to fall through a predetermined distance. The free body diagram of the oil drop is shown in Figure 9.7 (a), we note that viscous force and buoyant force balance the gravitational force.

Let the gravitational force acting on the oil drop (downward) be $F_g = mg$

Let us assume that oil drop to be spherical in shape. Let ρ be the density of the oil drop,

UNIT-9(XII-Physics_Vol-2)_New.indd 146 21-07-2020 15:48:25